



Understanding with the body? Testing the role of verb relative embodiment across tasks at the interface of language and memory

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ABSTRACT

Multiple representation accounts of conceptual knowledge argue that information associated with sensory-motor experience, in addition to pure linguistic information, contributes to word processing. A number of issues, however, remain under-investigated, including the extent to which these dimensions affect verb processing (rather than nouns), especially in languages other than English, and their role across different tasks along the language-memory continuum. Here, we collected ratings for a verb-specific dimension linked to bodily experience (*relative embodiment*, RE) for 647 Italian verbs and we tested its effects in three tasks differently modulating semantic activation and memory processes (i.e., lexical decision, grammatical decision, and memory recognition). Our results showed reliable influences of RE during lexical decision and memory recognition, but not in grammatical decision, possibly due to the Italian morphological richness. The cross-task analysis showed that RE effects were substantially higher in memory recognition compared to lexical decision, indicating that semantic and episodic processes interact at the interface of language and memory. Overall, results support the flexible and context-dependent role of sensory-motor and bodily-related experience during verb processing, pointing also to language-specific factors and implications for the organization of declarative memory.

Introduction

Through decades of debate within cognitive science, several attempts have been made to describe the nature of conceptual knowledge along a spectrum of proposals (for an updated summary, see Dove, 2023; Muraki et al., 2023). These range from purely amodal accounts, arguing that conceptual knowledge is essentially symbolic and detached from any specific modality (Fodor, 1975), to modal (or modality-specific) accounts put forward in the Embodied Cognition Framework, claiming that concepts are grounded in sensory-motor systems and that semantic processing corresponds to “re-enactments” or “simulations” of states involving the sensory-motor areas of the brain (Barsalou et al., 2003; Glenberg & Gallese, 2012).

In recent years, multiple representation theories have tried to move beyond the traditional opposition between modal and amodal accounts,

based on the idea that semantic knowledge is organized along different dimensions (Pexman, 2020), from amodal linguistic information (e.g., abstract semantic feature lists and semantic networks) to experience-based dimensions (e.g., sensory-motor, emotional, and social experience). The multidimensional nature of conceptual knowledge is primarily supported by empirical evidence reporting that lexical access is generally facilitated when a target word is associated with a richer semantic representation, involving not only higher involvement of sensorial and motor-related experience (Siakaluk et al., 2008) but also higher imageability (i.e., how easily a word can arouse mental images, see Yap et al., 2012), higher emotional experience (Siakaluk et al., 2016), and greater social relevance (Diveica, Muraki, Binney, & Pexman, 2024; Diveica, Pexman, & Binney, 2022).

Going further, multiple representation theories of conceptual knowledge predict that the involvement of amodal linguistic

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information vs. experience-based information varies as a function of the type and depth of processing activated by the task (Connell, 2019), as well as depending on individual differences (Pexman, 2020). In particular, a number of behavioral experiments reported significant, yet limited activation of sensory-motor information during tasks that require a shallower processing, such as lexical decisions (Yap et al., 2015; Yap, Pexman, et al., 2012), compared to tasks that involve deeper semantic processing, as in semantic decisions (Pexman et al., 2017). These task-related modulations of experience-based dimensions are generally explained by capitalizing on the interactive activation and competition (IAC) model (McClelland & Rumelhart, 1981), which is used to predict different patterns of top-down/bottom-up activations of orthographic, lexical, and semantic representations based on specific tasks (Balota, 1990). On the one hand, lexical decisions seem to rely more on lexical and orthographic properties of the stimuli, such as form familiarity, while semantic properties – including experience-based dimensions – affect performance only via feedback activations from semantic to orthographic representations (Pexman et al., 2002); on the other hand, semantic decisions require a complete semantic coding and are expected to involve higher activation of semantic-level properties compared to simple lexical decisions (Pexman et al., 2017). Second, studies on noun processing also documented a flexible interaction between experience-based dimensions and individual difference variables, such as vocabulary and mental imagery skills: in particular, individuals with higher vocabulary knowledge seem to capitalize more efficiently on the sensory-motor properties of the items in tasks that require deeper semantic processing (Pexman & Yap, 2018), while showing reduced sensitivity to word properties (including semantic ones) in tasks that require shallower processing (Yap, Pexman, et al., 2012). Conversely, individual differences in motor imagery skills interact with sensory-motor properties of the items only in very specific ways (Muraki, Dahm, et al., 2023; Muraki & Pexman, 2021).

However, the majority of the studies documenting effects related to task demands and individual differences focused on noun processing, while similar effects are less documented for verb processing. There is initial evidence that the activation of semantic properties affects verb processing too, with task and context modulations. For instance, Sidhu et al. (2014) analyzed a set of 687 English verbs belonging to different semantic classes (e.g., action, perception, cognitive processes, communication, social interactions, physiological states, etc.) and observed that verb processing was influenced by a novel semantic dimension, which they defined as *relative embodiment* (RE). RE expresses the experience related to “owning” and “sensing” the body (Borghi & Cimatti, 2010), including any kind of sensory-motor and bodily information deriving from voluntary actions, passive movements, proprioceptive experience, internal states, as well as from the interaction between the body and the environment. Interestingly, this study showed that during lexical decisions, RE played a greater role in responses than did verb imageability and was also a significant predictor of reaction times and accuracy in syntactic categorizations (i.e., decisions concerning grammatical categories of words), even when other lexical-semantic variables were controlled for, thus suggesting that bodily-related experience is particularly relevant to verb processing.

Another open issue concerns the extent to which the effect of experience-based dimensions spreads from lexical-semantic processing to other facets of language at the interface with memory skills (e.g., in word memory recognition). For instance, Sidhu and Pexman (2016) investigated the effect of the RE dimension on recognition memory for verbs, reporting that verbs associated with higher bodily experience were remembered more easily and more accurately compared to verbs less associated with bodily experience. Similarly, Muraki et al. (2022) reported that verbs characterized by higher RE ratings were easier to remember compared to various classes of abstract verbs (i.e., mental or emotional abstract verbs). This type of evidence seems to be in contrast with traditional theoretical views on declarative memory, which assume that semantic and episodic processes behave as distinct, segregated

memory components (Tulving, 1972). In fact, these sub-domains interact (for a recent overview, see De Brigard et al., 2022) and, as sparse data from verb processing seem to suggest, the boundaries between semantic and episodic domains become more permeable in tasks at the interface between language and memory.

Overall, across the last ten years a significant number of studies provided support to multiple representation accounts of semantic processing, showing the role of several dimensions and additional situational factors in word processing. However, only a small handful of studies addressed the role of task-dependent semantic modulation in verb processing, including the interaction between semantic and episodic processes at the language-memory interface. Importantly, the majority of empirical studies documented task-related influences on English verb processing only, while little evidence is available on the generalizability of such effects to other languages. This is a significant limitation for multiple representation accounts, given existing evidence that documents cross-linguistic variation of sensory-motor effects on language processing (Gianelli, Marzocchi, & Borghi, 2017; Kim, Müller, & Weiss, 2021).

The present study

The overarching aim of this study was to investigate a dimension related to bodily experience – namely RE – in a set of Italian verbs. In particular, we tested the effects of RE on verb processing across several tasks with different semantic loading along the language-memory interface.

To achieve our objective, we first acquired RE ratings for a large and heterogeneous set of Italian verbs, consistently with Sidhu et al. (2014), to test whether this experience-based dimension is relevant to Italian verbs. Then, we used the acquired RE ratings to test the role of this dimension on three visual word recognition paradigms with different levels of semantic demands, including lexical decision, grammatical decision (often referred to as *syntactic classification task*), and a memory recognition task involving both semantic and episodic processes. Overall, we expected to detect a flexible effect of RE during verb processing, consistent with multiple representation models; specifically, we expected RE effects to vary as semantic demands increase (up to involving episodic processes) and possibly due to language-specific factors.

To spell out the predictions in more detail, in the first place we relied on evidence collected for English. Specifically, we expected the effect of RE to be significant, yet moderate in a task with lower semantic demands, such as lexical decision. In particular, elaborating on evidence supporting the IAC model (McClelland & Rumelhart, 1981; see Yap et al., 2015), in this task we expected to observe larger effects of lexical-orthographic variables (i.e., lexical frequency, word length, and orthographic neighborhood, etc.) compared to experience-based semantic dimensions, such as RE. Going further, we predicted that, relative to shallower lexical decisions, tasks with higher semantic demands, such as grammatical decision, would trigger greater RE effects. The grammatical decision task is commonly used to test the involvement of conceptual properties of the stimuli at a deeper level of processing in English compared to lexical decisions (Yap et al., 2015), as the grammatical decision between verbs and nouns predominantly maps onto the more semantic “is it an action or an entity?” question, with the additional advantage of avoiding focusing on any particular semantic category as an explicit part of the task decision (Muraki, Dahm, et al., 2023; for a typological argument supporting verbs-as-actions vs. nouns-as-entities mapping, see Haspelmath, 2023; for a syntactic perspective on verb-noun asymmetry, see also Moro, 2020). This task is particularly appropriate for investigating verb representation and processing especially when combined with a go/no-go paradigm (i.e., with responses only to verbs), as that paradigm encourages more explicit processing of semantic dimensions related to verb meaning compared to a two-alternative forced-choice paradigm (Muraki et al., 2022). Data in English confirm these assumptions on the task by showing that

participants need to retrieve a full semantic coding of the word meaning to make their grammatical decisions, thus generating higher levels of activation for semantic-level proprieties, including those related to experience-based dimensions (Yap & Pexman, 2016). Capitalizing on this, we expected verbs with higher RE to be processed faster than verbs with lower RE, in line with prior evidence in English (Sidhu et al., 2014). Finally, the memory recognition task, in which participants have to decide whether a presented verb is either new or old (i.e., “old” items are those presented in a previous encoding task), is where we expected to observe the strongest effects of RE. Compared to previous tasks, the memory recognition task loads on different facets of declarative memory processes, including both lexical-semantic and episodic processing (Ballot et al., 2021; Cortese et al., 2010). Based on previous evidence from English verbs (see Muraki et al., 2022; Sidhu & Pexman, 2016), we then expected to see a significant generalizability of semantic-related facilitatory effects on episodic components of declarative memory (namely, better performance in the recognition of verbs with higher RE values compared to verbs with lower RE values). Importantly, as part of the multidimensional account, we also acknowledge possible language-specific effects (for a discussion, see Kemmerer, 2023) in determining the role of RE: while the above predictions are shaped on English data, the Italian material used here might disclose different patterns. The language-specific factor might become especially relevant when grammatical aspects are incorporated into the task (Bambini & Canal, 2021; Dillon & Keshev, submitted for publication), such as in the case of a grammatical decision task, where the rich morphological profile of Italian might reduce the semantic demands of the task and therefore exhibit a different involvement of semantic properties of the stimuli (see Vigliocco et al., 2005).

Finally, we included in the study an analysis of the role of individual difference variables in vocabulary and motor mental skills in relation with RE, as they might play a role in modulating its effect in verb processing (see Muraki & Pexman, 2021; Pexman & Yap, 2018; Yap, Pexman, et al., 2012). Given the scarcity of data in this domain, the analysis on this matter was conducted exploratorily.

Data availability

All data files, including novel RE rating values for Italian verbs, and the analysis codes for the present study are available through the Open Science Framework (OSF) at the following link: <https://osf.io/axhn3/>.

Rating task on Italian verbs

Method

Participants

One hundred Italian-speaking participants took part in the rating task. They were recruited via social media, public communications, and mailing lists. The sample was reduced to meet the distribution of participants' age reported in the English rating study reported in Sidhu et al. (2014), with a cut-off of $|2| SD$ (i.e., 26 years of age). Twenty-six participants were excluded as they were outside the range of 18–26 years of age.

Seventy-four participants were included in the final sample (43 female; Age, $M = 22.38$, $SD = 2.20$; Education in years, $M = 14.58$, $SD = 1.87$). Informed consent was obtained from all participants. The study (including the rating task and the other three experiments) was approved by the Ethics Committee of the Department of Brain and Behavioral Sciences of the University of Pavia (protocol number 76/2021).

Materials

The set of stimuli was constructed starting from the 687 English verbs used to collect RE ratings by Sidhu et al. (2014). We translated these verbs into Italian in their infinitive form following strict criteria: 1)

we selected the most immediate Italian equivalent according to the definitions provided by bilingual and monolingual dictionaries; 2) in the case of synonyms in the Italian translation, a lexical cognate (if present) was preferred over the other possible translations (e.g., eng. *to emit* it. ‘emettere’ instead of ‘emanare’); 3) in the case of verbs that function both in transitive and intransitive constructions in English and that in Italian can be translated either with or without the clitic pronoun *–si* (e.g., eng. *to comb* it. transitive ‘pettinare’, it. reflexive ‘pettinarsi’; eng. *to amuse* it. transitive ‘divertire’, it. intransitive ‘divertirsi’), we checked the most frequent use in English (based on the British National Corpus Online service, available on: <https://www.natcorp.ox.ac.uk/>) and we selected the Italian equivalent.

English items were excluded from the translation when: 1) the Italian translation would correspond to multi-word lexical units (e.g., eng. *to golf* it. ‘giocare a golf’, eng. *to yodel* it. ‘fare lo yodel’), or 2) a fitting translation was not available (e.g., *to braise*, *to cringe*, *to gargle*, *to scorch*). The English list also included near-synonymic pairs, that would lead to duplicate translations in Italian (e.g., eng. *to do/to make* > it. ‘fare’). In this case, only one English form was translated, choosing the closest in frequency to the Italian form. By these criteria, 30 verbs were excluded from the translation process.

The final translations were validated by comparing (log-transformed) lexical frequency values for English and Italian lemmas, extracted from Lund and Burgess (1996) and the “La Repubblica” corpus (Baroni et al., 2004), respectively. For four Italian verbs (.6% of the total set), log-transformed lexical frequency values were extracted from the itWaC corpus (Baroni et al., 2009), as they were not attested in the “La Repubblica” corpus. If a log-transformed value exceeded $|2.5| SD$ from the mean difference between English and Italian frequency values, we considered the pairs a case of lexical frequency mismatch. Nine verbs were excluded after this process (e.g., *to elapse*, *to echo*, *to need*, etc.). The remaining 648 Italian verbs showed a strong association with English lexical frequency values ($r(646) = .72$, $p < .001$) and were considered suitable for the rating task.

Procedure

The set of 648 Italian verbs was randomly split into two lists, each of which was divided into four blocks of equal length. The eight blocks were matched for log-transformed lexical frequency ($F(7, 640) = 0.60$, $p = .754$) and RE values associated with the English counterparts ($F(7, 640) = 0.15$, $p = .994$).

Thirty-five participants rated one list of verbs (“Survey A”), while 39 participants rated the other list (“Survey B”). To limit participants' fatigue, blocks within surveys were separated by short pauses. Randomization between and within blocks was used to prevent order-effect bias.

Both surveys were designed on the web-based platform LimeSurvey® (<https://limesurvey.org>). After providing informed consent and demographics (i.e., age, gender, years of education), participants received the instructions for the rating task (translated and adapted from Sidhu et al., 2014) and were asked to judge (on a 1–7 scale) the degree to which the meaning of each verb involved the human body. Each participant completed the survey autonomously as an online task.

Statistical analysis

We first assessed response quality with the intra-individual response variability score (IRV, Dunn et al., 2018) available in the *careless* package (Yentes & Wilhelm, 2023), computed within the entire set of responses and each block. Then, we tested rater reliability by computing the Interclass Correlation Coefficient (ICC), selecting a two-way model testing agreement on the average score with the *irr* package (Gamer et al., 2022). Finally, we described the distribution of Italian RE rating values and we validated them by testing their agreement with English RE values via Pearson's correlation. The analysis was performed using RStudio (R Core Team, 2023), version 2023.09.0+463.

Results

We obtained relative embodiment ratings for a total of 647 Italian verbs. One verb (it. *asserire* eng. ‘to assert’) was excluded after the rating task due to technical issues related to the online survey platform. Seven raters were excluded from the analysis because they obtained an IRV score < 1 in one of the four blocks. The ICC showed substantial to excellent agreement among the raters ($ICC_{SurveyA}=.88$, 95% CI [.86, .91], $p < .001$; $ICC_{SurveyB}=.93$, 95% CI [.92, .95], $p < .001$). During the rating task, all the participants used a range of points ≥ 6 on the 7-point scale, suggesting that the rating scale was used properly as instructed by the whole sample.

Italian RE values were distributed as follows: $M = 3.75$, $SD = 1.05$, $Mdn = 3.53$, $min = 2.08$, $max = 6.43$, and $range = 4.35$. The distribution showed a moderate positive skewness (0.64) and moderate negative kurtosis (-0.54). The distribution of Italian RE ratings is shown in Fig. 1A.

Italian and English RE ratings showed a strong positive correlation ($r(645) = .79$, $p < .001$), with a tight overlap between Italian and English RE rating distributions (see Fig. 1B).

The database with the rating values for Italian verbs, alongside the original English rating values, is openly accessible for future uses at the following link: <https://osf.io/axhn3/>.

Discussion

Our data from a heterogenous set of Italian verbs showed that bodily experience and the sense of “owning a body” varies considerably across verbs, with psychological verbs (e.g., it. ‘negare’ eng. *to deny*, it. ‘ponderare’ eng. *to ponder*, it. ‘presumere’ eng. *to assume*, etc.) receiving lower RE scores (RE values ranging between 2.08 and 2.19 in the verbs mentioned) and verbs related to motor and sensorial experience (e.g., it. ‘sedersi’ eng. *to sit*, it. ‘gustare’ eng. *to taste*, it. ‘calciare’ eng. *to kick*, etc.), physiological processes (e.g., it. ‘sanguinare’ eng. *to bleed*, it. ‘sudare’ eng. *to sweat*, it. ‘ingrassare’ eng. *to fatten*, etc.), and body interaction (e.g., it. ‘abbracciare’ eng. *to hug*, it. ‘coccolare’ eng. *to cuddle*, it. ‘lottare’ eng. *to fight*, etc.) receiving relatively higher RE scores (RE values ranging between 5.86 and 6.43 for the verbs mentioned). Importantly, our results show remarkable stability across raters, supporting the idea that sensory-motor, bodily, and interoceptive experience contribute to characterizing the semantic representation of verb meaning.

Finally, the cross-language correlation suggests that the extent to which speakers subjectively perceive bodily experience to be involved in

verb meaning is rather stable across languages. This finding expands previous evidence of cross-linguistic stability of other lexical-semantic variables, such as lexical frequency (Kuperman et al., 2024; Tjuka, 2020), imageability (Rofes et al., 2018), and emotional content (Marmolejo-Ramos et al., 2017).

Overall, the results of the rating task and the strong association with English values suggest that RE is a relevant semantic dimension for verbs across languages (Sidhu et al., 2014).

Experiment 1: Lexical decision task (LDT)

Method

Participants

We recruited 39 Italian-speaking young adult participants for the LDT. The sample size was decided following Sidhu et al. (2014), to have a comparable number of participants for this experiment. Exclusion criteria included being: 1) non-native speaker of Italian, 2) bilingual from birth, and 3) diagnosed with a learning disability. Four participants were excluded after checking for criteria 1) and 2). Thirty-five participants were included in the final sample (21 female; Age, $M = 22.85$, $SD = 2.85$; Education in years, $M = 14.49$, $SD = 2.01$). Informed consent was obtained from all participants (within ethical protocol 76/2021).

Materials

Following Sidhu et al. (2014), we selected 400 Italian verbs with relatively stable RE ratings, namely with low SD (the SD range in the resulting set was 1.04–1.88). The range of RE ratings in the Italian set was 2.08–6.43, spanning from low to high RE verbs. Focusing on our main hypothesis, namely that LDT item latencies would be affected by RE beyond other lexical variables, we ran an *a priori* power analysis with G*Power, version 3.1.9.6 (Faul et al., 2009), to determine the minimal set size of items. The power analysis indicated that 382 items were required to detect an effect size of $f^2 = .04$, as reported by Sidhu et al. (2014), with $\alpha = .05$ and power $\geq 80\%$.

A total of 400 pseudo-verbs (i.e., nonwords with a correct inflectional infinitival morpheme of Italian verbs, such as *febire*, with *-ire* as the infinitival marker) were created as control stimuli, matched with the 400 verbs for length ($t(798) = 0.16$, $p = .876$). For the verbs in the final set, we extracted a number of item variables, including log-transformed lexical frequency values from the “La Repubblica” corpus (Baroni et al., 2004), Orthographic Levenshtein Distance (OLD20) values, indicating word orthographic neighborhood, from the PhonItalia database (Goslin et al., 2013), the number of morphemes determined using the

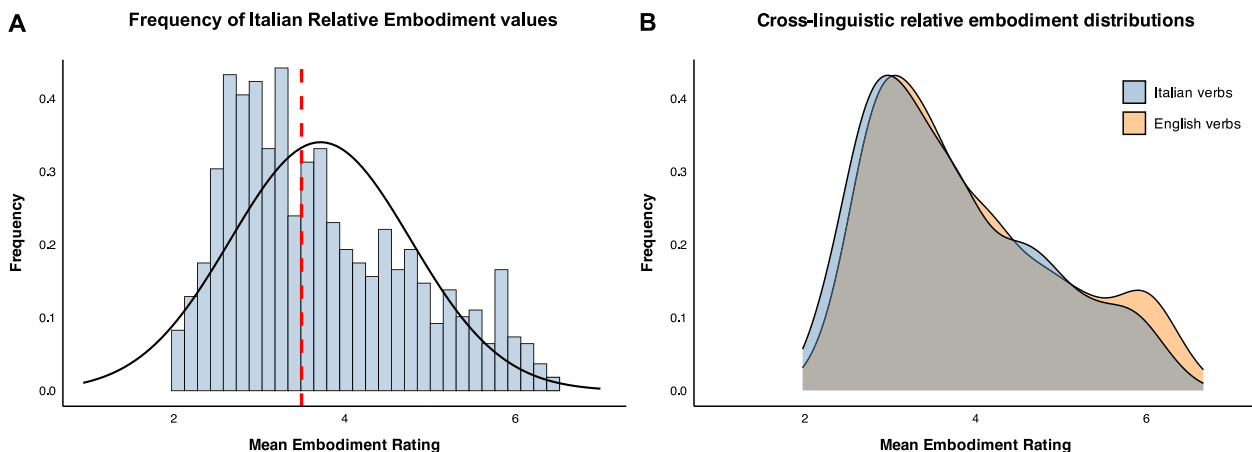


Fig. 1. Relative Embodiment rating values for Italian verbs. (A) The histogram displays the frequency of relative embodiment ratings for the 647 verbs used in the rating task (a black line representing a Gaussian distribution is plotted as a reference). The red dotted line represents the median of the distribution. (B) The density plots show the distributions of Italian and English relative embodiment ratings in comparison (the grey area represents the overlap region between the two distributions). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

DerIvaTario annotated lexicon (Talamo et al., 2016), and imageability values from the MEGAHR cross-linguistic repository (Ljubešić et al., 2018). The descriptive statistics of stimuli used for Experiment 1 are available in Table 1.

To exploratorily investigate the role of individual differences in motor mental imagery ability and vocabulary skills on LDT response, the Vividness of Movement Imagery Questionnaire-2 (VMIQ-2, Roberts et al., 2008) and the Verbal Meaning Task from the Primary Mental Abilities battery (Thurstone & Thurstone, 1963) were administered to participants as additional tasks.

The VMIQ-2 is a self-report questionnaire assessing the vividness of movement imagery. Participants were instructed to imagine 12 different actions from three different viewpoints (external visual imagery, internal visual imagery, and kinaesthetic imagery) on a 1–5 scale, with 1 corresponding to images that were perfectly clear and as vivid as normal vision or feeling of movement and 5 corresponding to no image at all. The questionnaire was adapted into Italian and implemented in a computerized format to be administered as an online task. To assess the reliability of the adapted version of the questionnaire, we tested its internal consistency with Cronbach’s α computed for External, Internal, and Kinaesthetic subscales (Cronbach’s $\alpha = .95, .96$, and $.94$ respectively). To improve interpretability, the VMIQ-2 scores across subscales were reversed before inclusion in the analysis, so that high VMIQ-2 scores indicate high motor imagery vividness.

The Verbal Meaning Task from the Primary Mental Abilities battery is a test measuring vocabulary skills, in particular, vocabulary depth (i. e., how well words are known and integrated in the mental lexicon, see Schmitt, 2014). Participants were instructed to identify the correct synonym of 50 target words by choosing from four given alternatives within 8 min. The total score ranged from 0 to 50. The Italian version of the Verbal Meaning Task was validated for in-person administration on a sample of young and older Italian-speaking adults (Cronbach’s $\alpha = .63$, see Rosi et al., 2017). Here, we implemented the Italian version of this task (Rosi et al., 2017) in a web-based computerized format to be administered online while maintaining the time pressure. Our online adaptation showed good internal consistency (Cronbach’s $\alpha = .80$, computed with Kuder-Richardson formula 20).

Cronbach’s α values for the VMIQ-2 and the Verbal Meaning Task were computed by pooling participants who completed Experiments 1, 2, and 3 ($N = 162$).

Procedure

Trials were divided into two blocks (each including 200 verbs and 200 pseudoverbs) matched for all the linguistic variables, separated by a short pause. Randomization between and within blocks was used to prevent order-effect bias.

Each trial began with a cross in the middle of the screen as a fixation point (lasting 1500 ms), followed by a target verb or a nonword that remained on the screen for 3000 ms. Participants were instructed to press the “L” key (or “A”, counterbalanced) on their keyboard using their right hand to respond “word” and the “A” key (or “L”, counterbalanced) using their left hand to respond “nonword”. Participants were asked to respond as quickly and as accurately as possible. Each participant completed 20 practice trials (10 verbs and 10 nonwords) before the experiment. Participants received feedback during the practice trials.

Table 1
Descriptive statistics (means and standard deviations) of the verbs used for Experiment 1.

Variable	Verbs ($N = 400$)
Relative Embodiment	3.76 (1.23)
Lexical Frequency (log)	7.18 (1.81)
Length (characters)	8.77 (1.61)
Orthographic Levenshtein Distance	2.16 (0.47)
Number of Morphemes	2.33 (0.54)
Imageability	3.57 (0.52)

After completing the LDT, participants were administered the VMIQ-2 and the Verbal Meaning Task within the same online session.

The experiment was implemented on the web-based platform Gorilla Experiment Builder® (<https://gorilla.sc>).

Statistical analysis

We first investigated the effect of item variables on LDT latencies, by computing patterns of associations between dimensions via a correlogram computed using Pearson’s correlations, adjusting p -values for False Discovery Rate (FDR, Benjamini & Hochberg, 1995). Then we assessed the effect of RE and other variables on LDT latencies by fitting a Linear Mixed-effects Model (LMM) on the reaction times of correct responses using *lme4* and *lmerTest* packages (Bates, Mächler, Bolker, & Walker, 2015; Kuznetsova, Brockhoff, & Christensen, 2017). Since reaction times distribution exhibited a positive skewness, we log-transformed the outcome variable to improve the normality of residuals and homogeneity of variance (Baayen, 2008). We also z-standardized log-transformed reaction times to allow for an easy comparison of the model’s estimates across experiments (Khanna & Cortese, 2021).¹

The LMM included item variables and, exploratorily, individual difference variables as fixed predictors. Item variables included two sets of dimensions: control (log-transformed lexical frequency, word length, OLD, and number of morphemes) and semantic variables (RE and imageability values). Individual difference variables included the VMIQ-2 subscale scores (External, Internal, and Kinaesthetic) and the Verbal Meaning Task total score.

Likelihood-ratio tests on models of increasing complexity were used to evaluate whether the inclusion of continuous predictors improved the model’s goodness of fit. We progressively added the control item variables to an intercept-only model, followed by semantic variables and individual differences variables, included in interaction with RE ratings. We kept in the model only those predictors that exerted a significant effect on the dependent variable. All continuous fixed predictors were standardized as z-scores before being included in the analysis. Inclusive R^2 (i.e., the total variance explained by a predictor) was computed for each fixed effect using the *PartR2* package (Stoffel, Nakagawa, & Schielzeth, 2021).

The optimal random structure was determined in a stepwise fashion using the *buildmer* package (Voeten, 2023), starting from a maximal model including all possible fixed and random effects. The final random structure included by-subjects and by-items random intercepts only, as more complex structures did not lead to model convergence. For each specified random effect, we checked the Best Linear Unbiased Predictors (Baayen, Davidson, & Bates, 2008): participants and items with systematic variation from the group mean (i.e., $> |2.5| SD$) were treated as deviant points and removed from the analysis.

The analysis was performed using RStudio (R Core Team, 2023).

Results

LDT data were pre-processed by trimming participants ($n = 0$) and items ($n = 8$) with accuracy values $< 70\%$. Incorrect trials and trials on which latencies were more than $|2.5| SD$ from the participant’s mean (2.91% of trials) were excluded from the analysis. Verb and nonword latencies and accuracy levels are reported in Table 2.

The correlogram showed that LDT log-transformed latencies for verbs were negatively correlated with log-transformed lexical frequency, RE, and Imageability, and positively correlated with word length, OLD, and number of morphemes (Fig. 2). Since Imageability and

¹ Additional analyses conducted for all mixed-effects models in Experiments 1–3 showed that all significant fixed effects remained regardless of the transformations applied to reaction times. The results of additional analyses are reported in the analysis code available in the OSF repository at the following link: <https://osf.io/axhn3/>.

Table 2

Descriptive statistics (means and standard deviations) for latencies (in milliseconds) and accuracy associated with verbs and nonwords in Experiment 1.

Variable	Mean	SD
LDT Latencies		
Verbs	722	65.42
Nonwords	849	76.10
LDT Accuracy		
Verbs	.96	.04
Nonwords	.96	.04

Note. LDT = Lexical Decision Task.

RE showed a high association and a similar pattern of correlations, we decided to drop Imageability (i.e., the least relevant variable considering our hypotheses) from subsequent analyses.

The best model included control variables, RE, and Verbal Meaning Task total score (Table 3). The model showed a significant negative effect of log-transformed lexical frequency and RE, as well as a positive effect of OLD on LDT latencies: these effects indicate that LDT latencies were faster in verbs with higher frequency and RE values but slower in verbs with a higher number of orthographic neighbours. Moreover, the model showed a significant interaction between RE and the Verbal Meaning Task total score, indicating that the effect of RE on latencies decreased as participants' vocabulary skills increased (see Table 4 and Fig. 3).

Discussion

The findings from the LDT on Italian verbs align with our prediction based on English data (Sidhu et al., 2014) and confirm the strong effect of lexical variables, on the one hand, and the modest effect of RE, on the other hand, in this task. First, the coupled effects of lexical frequency

and OLD support the idea that familiarity and orthographic processes occurring at the lexical level are the major drivers of performance in lexical decision. In particular, our data replicate the well-known word frequency (Gardner et al., 1987) and orthographic similarity (McClelland & Rumelhart, 1981) effects observed in LDTs, indicating that highly familiar words are processed faster, while words with a high number of orthographic neighbours are processed more slowly due to within-level lexical competition. Conversely, RE had only a modest effect on LDT latencies, as attested by the smaller portion of variance explained by this variable, confirming the idea that semantic properties facilitate word processing in lexical decisions via feedback activations to orthographic representations (Pexman et al., 2002).

Interestingly, the results also confirm our predictions concerning individual difference variables. In particular, we observed a significant interaction between verb RE values and participants' vocabulary skills: the magnitude of the facilitatory effect of RE on latencies decreased as participants' vocabulary skills increased. Conversely, we did not observe any effect of motor mental imagery skills. These findings are in line with the idea that higher vocabulary skills (but not imagery) facilitate lexical decisions (Muraki & Pexman, 2021), with individuals with higher vocabulary skills relying less on semantic dimensions of words (such as RE) to make lexical decisions, compared to less lexically skilled participants (see also Yap, Pexman, et al., 2012).

Experiment 2: Grammatical decision task (GDT)

Method

Participants

For the GDT, we ran an *a priori* power analysis based on our major research question, namely whether participants would process verbs with higher RE better compared to lower RE verbs. The power analysis

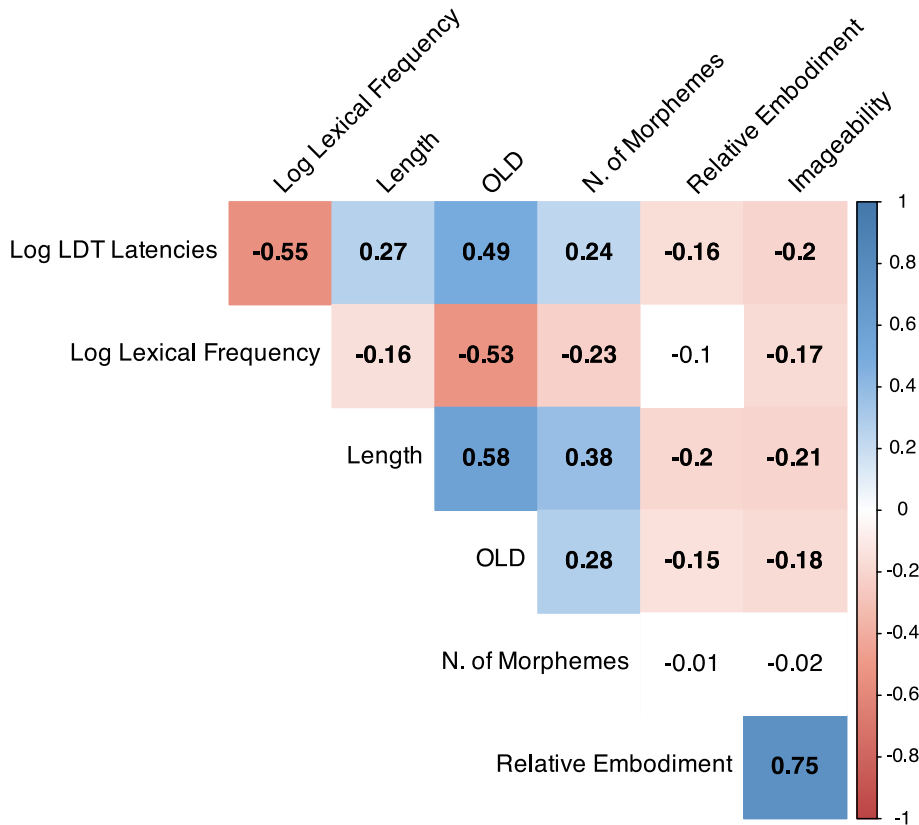


Fig. 2. Correlogram between verb LDT latencies and item variables. The magnitude of associations is depicted by colour (white cells indicate non-significant correlations after FDR correction, with significance level $p < .05$). Note: LDT = Lexical Decision Task; OLD = Orthographic Levenshtein Distance.

Table 3

Likelihood-ratio tests to assess the goodness of fit of the LMMs on LDT log-transformed zRTs, based on the progressive inclusion of fixed effects.

Models' fixed effects	AIC	BIC	logLik	χ^2	df	p-value
Control variables	25,463	25,522	−12,724	172.34	4	<.001
Control variables + RE	25,450	25,515	−12,716	15.51	1	<.001
Control variables + RE * (VMIQ-2 Ext + VMIQ-2 Int + VMIQ-2 Kin)	25,457	25,566	−12,714	4.91	6	.555
Control variables + RE * Verbal Meaning Task	25,443	25,523	−12,710	11.00	2	.004

Note. RE = Relative Embodiment; VMIQ-2 = Vividness of Movement Imagery Questionnaire 2; Ext = External; Int = Internal; Kin = Kinaesthetic. Control variables include log-transformed lexical frequency, word length, Orthographic Levenshtein Distance, and number of morphemes.

indicated that 47 participants were required to detect an effect size of Cohen's $d = 0.42$ with $\alpha = .05$ and power = 80%, as reported by Sidhu et al. (2014) for the same experiment, which was increased by $\geq 10\%$ to account for exclusions. We recruited 53 Italian-speaking young adult participants who did not take part in Experiment 1. Exclusion criteria included being: 1) non-native speaker of Italian, 2) bilingual from birth, and 3) diagnosed with a learning disability. Nine participants were excluded after checking for these criteria. Forty-four participants were included in the final sample (35 female; Age, $M = 23.91$, $SD = 3.21$; Education in years, $M = 15.11$, $SD = 2.05$). A sensitivity power analysis with $\alpha = .05$ and power = 80% confirmed that our sample size was sufficient to detect effect sizes greater than Cohen's $d \geq 0.43$. Informed consent was obtained from all participants (within the ethical protocol 76/2021).

Materials

Following Sidhu et al. (2014), we selected 80 Italian verbs from the set of stimuli used in the rating task and also 80 Italian nouns. Since in Italian infinitival forms are marked by specific morphological suffixes (i. e., *−are*, *−ere*, *−ire*) which are not present in nouns, all verbs were presented in the 3SG present form (i. e., with the inflectional marking *−a* or *−e*, as in it. *mangi-a*, eng. 'eats.3SG'), i. e., the most frequent form of most Italian verbs (Bertinetto et al., 2020). Italian verbs in the 3SG present form inflectional marking might result in homophones with nouns, so the set of items was adjusted in order to avoid verbs that could also be homophone nouns (e. g., it. *cucina* 'kitchen' but also 'cooks.3SG'). We also substituted verbs with the clitic pronoun *−si* in the infinitival form resulting in multiword expressions in the 3SG present form (e. g., it. *allontanarsi* 'to depart' > *si allontana* 'departs.3SG'). Finally, to avoid orthographic confounds, we excluded nouns with endings not represented in the 3SG present form of verbs (i. e., *−o*, since 3SG verbs can end only with *−a* and *−e* in Italian). The final list of verbs was constructed to have two subsets of items, each including 40 verbs, characterized by

high (> 3.5) and low (< 3.5) RE ratings. Using a dichotomized variable for RE is consistent with Sidhu et al. (2014) and other studies along the same lines (Muraki, Dahm, et al., 2023; Muraki & Pexman, 2021; Sidhu & Pexman, 2016) and allows us to easily control for all potential confounding variables. The two verb subsets were matched for relevant lexical and semantic variables, such as log-transformed lexical frequency, word length, Orthographic Levenshtein Distance (OLD20), number of morphemes, and imageability ($ts < 0.98$, $ps > .331$), and they were significantly different only for RE values ($t(78) = 11.58$, $p < .001$). The list of verbs selected for the GDT was matched also with the set of nouns for lexical variables and imageability ($ts < 1.41$, $ps > .160$, see Table 5).

Procedure

Trials ($N = 160$) were administered to participants in two separate blocks (each including 40 verbs and 40 nouns), matched for all the relevant dimensions. Randomization was applied within and between blocks. Blocks were separated by a short pause. Each trial began with a fixation cross for 1000 ms, followed by a blank screen for 1000 ms. Afterwards, a target verb or a noun was presented. The task was combined with a *go/no-go* paradigm, in which participants were instructed to decide whether each word was a verb or a noun. If they judged that the word was a verb, they had to press the space key on their keyboard. Otherwise, they were instructed to refrain from responding. The next trial began automatically after 3000 ms. Each participant completed 14 practice trials (7 verbs and 7 nouns) before starting the experiment. During the practice trials, participants received feedback. The experiment was administered on the web-based platform Gorilla Experiment Builder®. After completing the GDT, participants were administered the VMIQ-2 and the Verbal Meaning Task within the same online session.

Statistical analysis

Following a similar analytical strategy to the one adopted for

Table 4

Output of the best Linear Mixed-effects Model on LDT log-transformed zRTs.

Fixed Effects	β	95% CI	p-value	IR ²
(Intercept)	−0.05	[−0.22, 0.12]	.562	–
Lexical Frequency (log)	−0.11	[−0.14, −0.09]	<.001	.023
Word Length	−0.00	[−0.03, 0.02]	.833	.005
Orthographic Levenshtein Distance	0.06	[0.03, 0.09]	<.001	.019
N. of morphemes	0.02	[−0.00, 0.04]	.076	.004
Relative Embodiment	−0.04	[−0.06, −0.02]	<.001	.002
Verbal Meaning Task	−0.09	[−0.26, 0.08]	.285	.009
Relative Embodiment * Verbal Meaning Task	0.02	[0.01, 0.04]	.006	<.001
Random Effects	Variance	SD		
Intercept _{Subject}	0.25	0.50		
Intercept _{Item}	0.02	0.13		
Residuals	0.60	0.77		
ICC _{SubjectItem}	0.31			
Model fit	Marginal	Conditional		
R ²	.040	.337		

Note. LDT = Lexical Decision Task; β = model estimates (standardized); CI = confidence intervals; IR² = Inclusive R².

Only complete cases entered the model. One deviant subject was excluded from the analysis. Model formula: Log-transformed zRT ~ Log Lexical Frequency + Length + OLD + N. of morphemes + Relative Embodiment * Verbal Meaning Task + (1 | Subject) + (1 | Item).

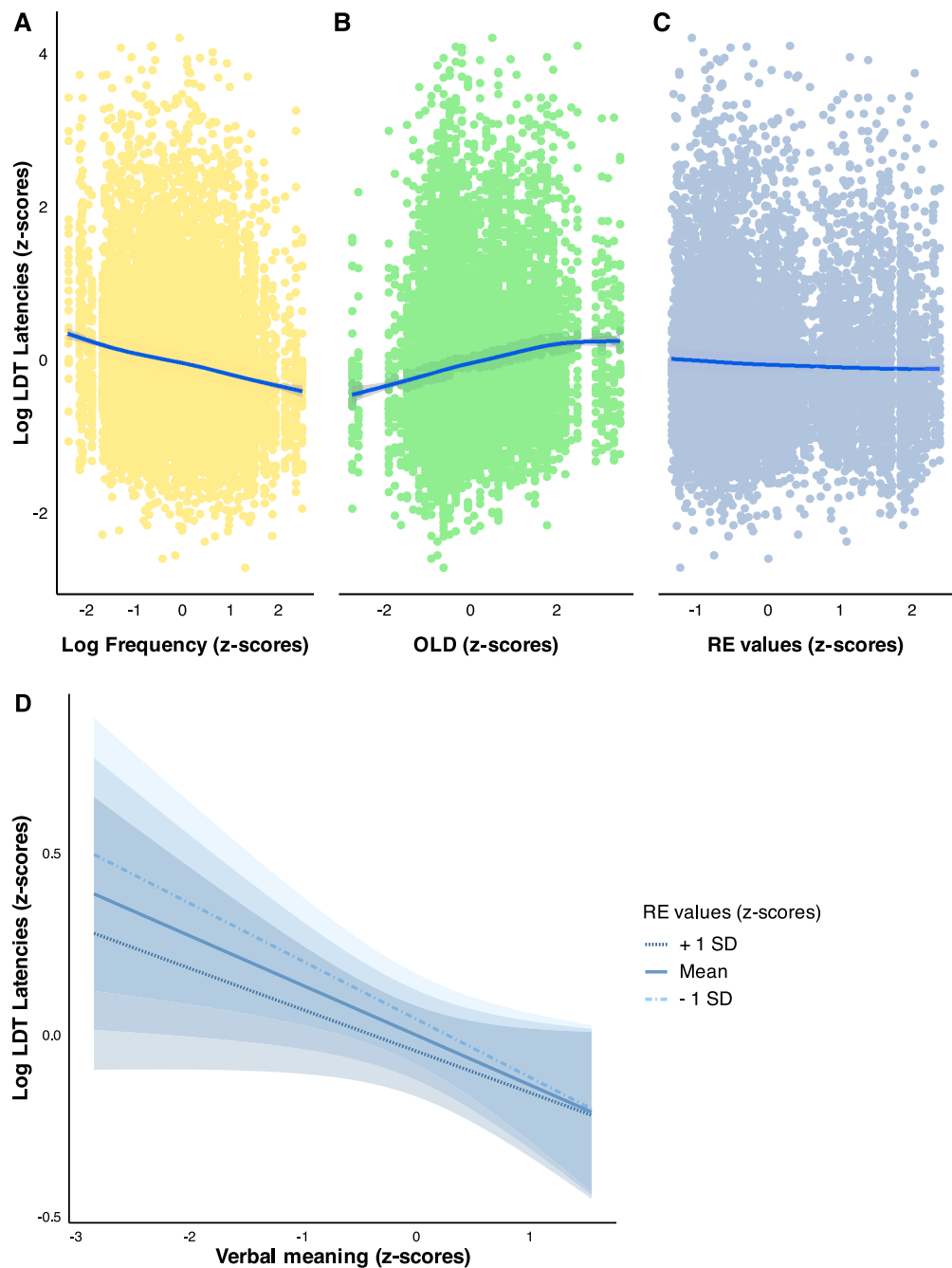


Fig. 3. Significant effects of item and individual difference variables on lexical decision latencies. (A-C) Fitted values for log-transformed z-scored Lexical Decision Task (LDT) latencies predicted by significant fixed-effects variables, namely log-transformed lexical frequency values, Orthographic Levenshtein Distance (OLD), and Relative Embodiment (RE) values (smoothed lines added to aid visualization). (D) Fitted values for log-transformed z-scored LDT latencies predicted by RE values in interaction with Verbal Meaning Task total score.

Experiment 1 (see Section 4.1.4), we fit a Linear Mixed-effects Model (LMM) on the log-transformed z-scored reaction times of correct responses. The LMM included RE levels (low vs. high, sum contrast coded, baseline = low) and individual differences variables, namely VMIQ-2 reversed subscale scores (External, Internal, and Kinaesthetic) and the Verbal Meaning Task total score.

Likelihood-ratio tests were again used to evaluate the inclusion of standardized continuous predictors, starting from RE levels added to an intercept-only model, followed by individual differences variables in interaction with RE levels.

The final random structure included by-subjects and by-items random intercepts only, as the optimal structure (including by-items

random slope for the Verbal Meaning task total score) led to convergence issues when individual differences variables were included in the model.

The analysis was performed using RStudio (R Core Team, 2023).

Results

GDT data were pre-processed by trimming participants ($n = 1$) with accuracy values $< 70\%$. Incorrect trials and trials on which latencies exceeded $|2.5|$ SD from the participant's mean were excluded from the analysis (3.24% of trials). Descriptive statistics for by-subject and by-item latencies and accuracy are reported in Table 6.

Table 5

Descriptive statistics (means and standard deviations) of the verbs used for Experiment 2 (low and high Relative Embodiment verbs and the matched control nouns).

Variable	Low RE verbs (N = 40)	High RE verbs (N = 40)	Nouns (N = 80)
Relative Embodiment ^a	2.75 (0.40)	4.46 (0.84)	–
Lexical Frequency (log) ^a	7.94 (1.39)	7.93 (1.72)	8.11 (1.70)
Length (characters) ^b	6.93 (1.54)	7.10 (1.58)	7.10 (1.50)
Orthographic Levenshtein Distance ^b	1.91 (0.35)	2.00 (0.42)	2.04 (0.40)
Number of Morphemes ^b	2.23 (0.42)	2.25 (0.44)	2.21 (0.41)
Imageability ^a	3.54 (0.39)	3.63 (0.46)	3.65 (0.44)

Note. RE = Relative Embodiment.

^avalue of the lemma (infinitive form) used for verbs

^bvalue of the form (3SG present indicative) used for verbs

Table 6

Descriptive statistics (mean and standard deviations) of by-subject and by-item latencies (in milliseconds) and accuracy for low and high relative embodiment verbs.

Measure	Low RE	High RE
GDT Latencies		
By-subject	783 (221)	781 (201)
By-item	778 (48.60)	780 (64.90)
GDT Accuracy		
By-subject	.99 (.04)	.99 (.03)
By-item	.99 (.02)	.99 (.02)

Note. RE = Relative Embodiment; GDT = Grammatical Decision Task.

Table 7

Likelihood-ratio tests to assess the goodness of fit of the LMMs on GDT log-transformed zRTs, based on the progressive inclusion of fixed effects.

Models' fixed effects	AIC	BIC	logLik	χ^2	df	p-value
RE levels	7,317.2	7,347.7	–3,653.6	0.07	1	.798
VMIQ-2 Ext + VMIQ-2 Int + VMIQ-2 Kin	7,321.2	7,363.8	–3,653.6	0.09	3	.993
Verbal Meaning Task	7,295.1	7,325.6	–3,642.6	22.12	1	<.001

Note. RE = Relative Embodiment; VMIQ-2 = Vividness of Movement Imagery Questionnaire 2; Ext = External; Int = Internal; Kin = Kinaesthetic.

The hierarchical approach testing the effect of fixed predictions showed that the inclusion of RE levels and VMIQ-2 subscales did not improve model's fit.² Only the inclusion of the Verbal Meaning Task total score showed a main effect, indicating that GDT latencies were generally faster for individuals with higher vocabulary skills. The output of the Likelihood-ratio tests and the best model are reported in Tables 7 and 8.

Discussion

The results of the GDT did not align with the English-oriented prediction of strong activation of semantic properties when participants had to decide whether a word was a noun or a verb. English speakers were shown to exploit high activation of verb semantic properties in this task, resulting in faster and better responses for high RE verbs compared to low RE ones (Sidhu et al., 2014). Conversely, RE was not a significant predictor of GDT response in Italian, a language in which verb-noun class assignment does not seem to require semantic processing (see Postiglione & Laudanna, 2016). The pattern is in line with multiple

² We conducted an additional analysis to check whether the inclusion of RE as a continuous variable would change the results of the GDT. This additional check confirmed that RE does not affect GDT latencies even when included as a continuous variable ($\beta = 0.01$, $SE = 0.02$, $t = 0.25$, $p = .801$). The results are reported in the analysis code available in the OSF repository at the following link: <https://osf.io/axhn3/>.

Table 8

Output of the best Linear Mixed-effects Model on GDT log-transformed zRTs.

Fixed Effects	β	95% CI	p-value	IR ²
(Intercept)	–0.03	[–0.19, 0.13]	.698	–
Verbal Meaning Task	–0.31	[–0.48, –0.14]	.001	.095
Random Effects	Variance	SD		
Intercept _{Subject}	0.24	0.49		
Intercept _{Item}	0.03	0.18		
Residuals	0.48	0.70		
ICC _{SubjectItem}	0.36			
Model fit	Marginal	Conditional		
R ²	.095	.418		

Note. GDT = Grammatical Decision Task; β = model estimates (standardized); CI = confidence intervals; IR² = Inclusive R². One deviant subject was excluded from the analysis. Model formula: Log-transformed zRT ~ Verbal Meaning Task + (1 | Subject) + (1 | Item).

representation models and shows that the activation of sensory-motor properties of words is influenced by language-specific factors, such as differences at the grammatical level (Gianelli et al., 2017). In particular, this pattern suggests that in a richly inflected language such as Italian, word morphological representation interferes with other levels of processing (Bambini & Canal, 2021). In other words, in a task where judgments over grammatical categories are required, the performance of Italian-speaking individuals might be driven by morphological information embedded in Italian verbs and nouns, rather than semantic properties.

As for individual differences, the only significant predictor for GDT latencies was vocabulary skills: this effect is in line with previous evidence showing that participants with deeper vocabulary knowledge and higher integration of words within the mental lexicon capitalize on more extended and robust word knowledge, resulting in faster grammatical decisions compared to less skilled participants (see Muraki & Pexman, 2021).

Experiment 3: Memory recognition task (MRT)

Method

Participants

For the MRT, we ran an *a priori* power analysis based on the hypothesis that verbs with higher RE would be recognized better than lower RE verbs. The power analysis indicated that 75 participants were required to detect an effect size of Cohen's $d = 0.33$ with $\alpha = .05$ and power = 80%, reported by Sidhu & Pexman (2016) for the same experiment administered online, which was increased by $\geq 10\%$ to account for exclusions. Ninety-eight Italian-speaking young adult participants who did not take part in Experiments 1 and 2 were involved in the MRT. Exclusion criteria included being: 1) non-native speaker of Italian, 2) bilingual from birth, and 3) diagnosed with a learning disability. A total of 13 participants were excluded after checking for these criteria.

Table 9

Descriptive statistics (means and standard deviations) of the verbs used for Experiment 3 (low and high Relative Embodiment verbs for lists 1 and 2 with the matched control nouns).

Variable	List 1 (N = 70)		List 2 (N = 70)		Nouns (N = 70)
	Low RE verbs	High RE verbs	Low RE verbs	High RE verbs	
Relative Embodiment ^a	2.85 (0.37)	4.46 (0.76)	2.81 (0.36)	4.44 (0.80)	—
Lexical Frequency (log) ^a	7.85 (1.48)	7.89 (1.58)	8.08 (1.37)	7.95 (1.68)	8.00 (1.69)
Length (characters) ^b	6.89 (1.55)	6.71 (1.66)	7.14 (1.50)	6.80 (1.49)	6.99 (1.49)
OLD ^b	1.92 (0.34)	1.90 (0.33)	1.94 (0.41)	1.96 (0.44)	1.99 (0.36)
Number of Morphemes ^b	2.26 (0.44)	2.31 (0.47)	2.23 (0.43)	2.23 (0.43)	2.21 (0.41)
Imageability ^a	3.59 (0.44)	3.67 (0.44)	3.53 (0.31)	3.62 (0.44)	3.68 (0.43)

Note. RE = Relative Embodiment; OLD = Orthographic Levenshtein Distance.

^avalue of the lemma (infinitive form) used for verbs

^bvalue of the form (3SG present indicative) used for verbs

Table 10

Descriptive statistics (mean and standard deviations) and comparisons of participants' *d*-prime scores, hits rates, and false alarm rates for low and high relative embodiment verbs.

Variable	Low RE	High RE	<i>t</i> -statistics	Cohen's <i>d</i>	<i>p</i> -value
<i>D</i> -prime	0.92 (0.46)	1.07 (0.48)	<i>t</i> (73) = 2.90	0.34	.005
Hit rate	0.67 (0.08)	0.70 (0.09)	<i>t</i> (73) = 2.42	0.28	.018
False alarm rate	0.33 (0.08)	0.39 (0.09)	<i>t</i> (73) = -2.42	-0.28	.018

Note. RE: Relative Embodiment.

Eighty-five participants were included in the final sample (48 female; Age, $M = 25.56$, $SD = 4.14$; Education, $M = 16.34$, $SD = 2.46$). A sensitivity power analysis with $\alpha = .05$ and power = 80% confirmed that our sample was sufficient to detect effect sizes greater than Cohen's $d \geq .31$.

Informed consent was obtained from all participants (within the ethic protocol 76/2021).

Materials

Consistent with Sidhu and Pexman (2016), we selected 140 Italian verbs, equally divided into two lists, and a single list of 70 nouns. For this task, we expanded the set of verbs from Experiment 2, by adding 60 verbs from the list in the rating task. Verbs were again presented in their 3SG present indicative form, therefore the selection of the additional items followed the same criteria used in Experiment 2 (see Section 5.1.2). Each verb list included 35 verbs with high RE values and 35 verbs with low RE values, selected using the same cut-off as in Experiment 2 (low: $RE < 3.5$; high: $RE > 3.5$). The verb lists were matched for log-transformed lexical frequency, word length, Orthographic Levenshtein Distance (OLD), number of morphemes, imageability, and RE values ($ts < 0.77$, $ps > .443$). Both verb lists were matched for the same variables (excluding RE) with the list of nouns ($Fs < 0.99$, $ps > .370$, see Table 9).

Procedure

In this experiment, participants first took part in a *go/no-go* GDT (with the same procedure as in Experiment 2), during which they were presented with only one list of verbs and the list of nouns for a total of 140 trials (i.e., 70 verb trials from either list 1 or 2 and 70 noun trials). Participants were all unaware that their verb memory would be tested in a later task. Verb list selection for the GDT was counterbalanced across participants and combined with the single list of nouns. Afterwards, participants took part in the MRT including both lists of verbs (140 trials) presented in a randomized order, in which they had to decide whether a verb on the screen had already been presented to them in the previous task (i.e., an *old* verb) or not (i.e., a *new* verb). Each trial began with a cross in the middle of the screen as a fixation point (lasting 1000 ms), followed by a verb that remained on the screen for 3000 ms. Participants were instructed to press the "L" key (or "A", counterbalanced)

on their keyboard using their right hand to respond "old" and the "A" key (or "L", counterbalanced) using their left hand to respond "new". The next trial began automatically after 3000 ms if no answer was provided. The MRT was preceded by a five-minute distraction task, in which participants had to answer as many addition questions (e.g., "364 + 133 = ?") as they could. The experiment was administered on the web-based platform Gorilla Experiment Builder®. After completing the MRT, participants were administered the VMIQ-2 and the Verbal Meaning Task within the same online session.

Statistical analysis

Building on the classic approaches to the analysis of memory recognition data, we first computed *d*-prime scores to compare participants' performance with low-RE verbs and high-RE verbs. *D*-prime scores were computed using the *psycho* package (Makowski, 2018) as: *z*-scored hit rate (i.e., percent of trials on which a previously seen word was correctly identified as an old word) minus *z*-scored false alarm rate (i.e., percent of trials on which a new word was incorrectly identified as an old one). Participants obtaining negative *d*-prime scores were flagged as having poor reliability and removed from the analysis, as they might reflect sampling error or response confusion (Stanislaw & Todorov, 1999). Then, we compared participants' *d*-prime scores, hit rates, and false alarms using paired *t*-tests.

Afterwards, we fitted a Linear Mixed-effects Model (LMM) on the log-transformed *z*-scored reaction times of the correct answers of the MRT, following the same criteria used in Experiments 1 and 2 (see Sections 4.1.4 and 5.1.4). The optimal random structure for model convergence included by-subjects and by-items random intercepts and a by-subjects random slope for RE levels.

The analysis was performed using RStudio (R Core Team, 2023).

Results

Memory recognition task results

We first inspected data quality to trim participants who were not appropriately committed during the entire experiment. The exclusion criteria were: a) $< 70\%$ correct answers in either critical or control trials of the encoding task ($n = 0$), b) < 10 additions in the distraction task (n

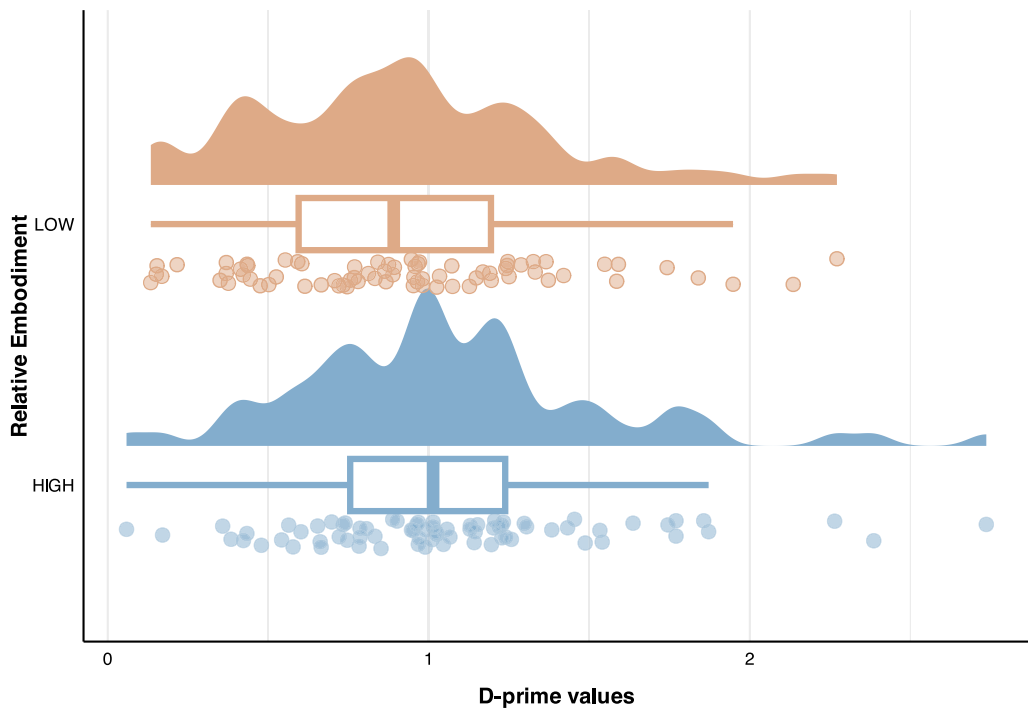


Fig. 4. Accuracy values in the Memory Recognition task. The raincloud plots show the distribution of *d*-prime scores obtained by participants for high-RE and low-RE verbs.

Table 11
Likelihood-ratio tests to assess the goodness of fit of the LMMs on MRT log-transformed *z*RTs, based on the progressive inclusion of fixed effects.

Models' fixed effects	AIC	BIC	logLik	χ^2	df	<i>p</i> -value
RE levels	17,406	17,454	−8,696.1	4.20	1	.040
RE levels * (VMIQ-2 Ext + VMIQ-2 Int + VMIQ-2 Kin)	17,413	17,501	−8,693.4	5.44	6	.488
RE levels * Verbal Meaning Task	17,404	17,465	−8,693.0	6.23	2	.044

Note. RE = Relative Embodiment; VMIQ-2 = Vividness of Movement Imagery Questionnaire 2; Ext = External; Int = Internal; Kin = Kinaesthetic.

= 2), c) > 20 min between the end of the encoding task and the end of the MRT (*n* = 7), d) > 30% of time-out trials in the MRT (*n* = 2), and e) obtaining negative *d*-prime scores (*n* = 3). To analyze latencies, we trimmed incorrect trials and trials on which latencies were more than |

Table 12
Output of the best Linear Mixed-effects Model on MRT log-transformed *z*RTs for all correct answers.

Fixed Effects	β	95% CI	<i>p</i> -value	IR ²
(Intercept)	0.01	[−0.09, 0.11]	.830	—
RE levels (Low vs. High)	−0.08	[−0.16, −0.00]	.038	.002
Verbal Meaning Task	−0.13	[−0.22, −0.03]	.009	.016
RE levels (Low vs. High) * Verbal Meaning Task	−0.03	[−0.10, 0.03]	.320	<.001
Random Effects				
Intercept _{Subject}	0.17	0.41		
RE levels (Low vs. High) _{Subject}	0.04	0.21	−.35	
Intercept _{Item}	0.02	0.13		
Residuals	0.76	0.87		
ICC _{SubjectItem}	0.20			
Model fit	Marginal	Conditional		
R ²	.018	.218		

Note. MRT = Memory Recognition Task; β = model estimates (standardized); CI = confidence intervals; IR² = Inclusive R². One deviant subject was excluded from the analysis. Model formula: Log-transformed *z*RT ~ RE Levels * Verbal Meaning Task + (1 + RE Levels | Subject) + (1 | Item).

2.5| *SD* from the participant's mean (3.70% of trials).

Paired *t*-tests on participants' *d*-primes, hit rates, and false alarm rates showed that verbs with high RE ratings were more accurately recognized than verbs with low RE ratings (Table 10, Fig. 4).

As for the LMM on MRT latencies, the inclusion of RE levels and Verbal Meaning Task total scores to the baseline model explained most data variance: both variables showed a main effect, indicating that MRT latencies were faster for high RE verbs and in individuals with higher vocabulary skills.³ Conversely, no effects were observed for motor mental imagery (see Table 11 for the output of the Likelihood-ratio tests and Table 12 for the output of the best model).

Post-hoc cross-task comparisons

After investigating the role of RE in different tasks, we aimed to exploratorily compare its involvement across tasks, by comparing the results obtained in the two tasks where it exerted a significant effect, namely LDT (Experiment 1) and MRT (Experiment 3).

To achieve this objective, we compared trimmed log-transformed *z*-scored latencies and binomial Accuracy scores (0 = incorrect, 1 =

³ We conducted an additional analysis to check whether the inclusion of RE as a continuous variable would change the results of the MRT. This additional check confirmed that higher RE levels correspond to faster responses in the MRT even when RE is included as a continuous variable ($\beta = -0.06$, *SE* = 0.01, *t* = −4.30, *p* < .001). The results are reported in the analysis code available in the OSF repository at the following link: <https://osf.io/axhn3/>.

Table 13

Output of the Linear Mixed-effects Model on LDT and MRT log-transformed zRTs for all correct answers.

Fixed Effects	β	95% CI	p-value	IR ²
(Intercept)	−0.02	[−0.12, 0.08]	.702	–
Lexical Frequency (log)	0.01	[−0.03, 0.04]	.699	.000
Word Length	0.04	[−0.00, 0.08]	.074	.004
Orthographic Levenshtein Distance	0.02	[−0.01, 0.06]	.256	.003
N. of morphemes	0.02	[−0.01, 0.04]	.252	.001
Verbal Meaning Task	−0.10	[−0.19, −0.01]	.033	.011
Relative Embodiment	−0.04	[−0.06, −0.01]	.006	.002
Task (LDT vs. MRT)	−0.02	[−0.22, 0.18]	.839	.022
Relative Embodiment * Task (LDT vs. MRT)	0.08	[0.04, 0.12]	<.001	.005
Random Effects	Variance	SD	Corr	
Intercept _{Subject}	0.22	0.47		
Lexical Frequency (log) _{Subject}	0.01	0.11	.27	
Intercept _{Item}	0.00	0.07		
Residuals	0.71	0.84		
ICC _{SubjectItem}	0.25			
Model fit	Marginal	Conditional		
R ²	.017	.263		

Note. LDT = Lexical Decision Task; MRT = Memory Recognition Task; β = model estimates (standardized); CI=confidence intervals; IR² = Inclusive R². One deviant subject was excluded from the analysis. Model formula: Log-transformed zRT ~ Log Lexical Frequency + Length + OLD + N. of morphemes + Verbal Meaning Task + Relative Embodiment * Task + (1 + Log Lexical Frequency | Subject) + (1 | Item).

correct) obtained in the LDT and the MRT by fitting a LMM and a Generalized Linear Mixed-effects Model (GLMM), respectively. For this analysis, we included latencies and accuracy obtained by participants in a subset of verbs present in both experiments ($n = 83$ verbs, 21% of the LDT set and 59% of the MRT set). We included Task (LDT vs. MRT, with sum contrast coding and MRT as baseline) in interaction with RE ratings as fixed predictors. Based on previous analysis, showing a significant

effect of specific item dimensions and vocabulary skills, we also included control item variables (i.e., log-transformed lexical frequency, word length, OLD, and the number of morphemes) and Verbal Meaning Task total score as covariates. The LMM random structure included by-subjects and by-items random intercepts and by-subjects random slope for log-transformed lexical frequency, while the GLMM random structure included by-subjects intercepts only. The analysis was performed using RStudio (R Core Team, 2023) with packages already mentioned in Section 4.1.4.

Table 14

Output of the Generalized Linear Mixed-effects Model on LDT and MRT binomial Accuracy scores.

Fixed Effects	OR	95% CI	p-value	IR ²
(Intercept)	11.04	[9.46, 12.88]	<.001	–
Lexical Frequency (log)	0.86	[0.81, 0.91]	<.001	.002
Word Length	0.83	[0.75, 0.92]	<.001	.056
Orthographic Levenshtein Distance	1.05	[0.96, 1.15]	.267	.010
N. of morphemes	0.96	[0.91, 1.02]	.219	<.001
Verbal Meaning Task	1.01	[0.94, 1.09]	.770	.017
Relative Embodiment	0.99	[0.86, 1.14]	.889	<.001
Task (LDT vs. MRT)	36.11	[26.22, 49.71]	<.001	.310
Relative Embodiment * Task (LDT vs. MRT)	0.85	[0.64, 1.12]	.249	.001
Random Effects	Variance	SD		
Intercept _{Subject}	0.07	3.29		
ICC _{SubjectItem}	0.02			
Model fit	Marginal	Conditional		
R ²	.391	.435		

Note. LDT = Lexical Decision Task; MRT = Memory Recognition Task; OR = odds ratios; CI = confidence intervals; IR² = Inclusive R². Model diagnostics were inspected using DHARMa R package (Hartig, 2022). One deviant subject was excluded from the analysis. Model formula: Accuracy ~ Log Lexical Frequency + Length + OLD + N. of morphemes + Verbal Meaning Task + Relative Embodiment * Task + (1 | Subject).

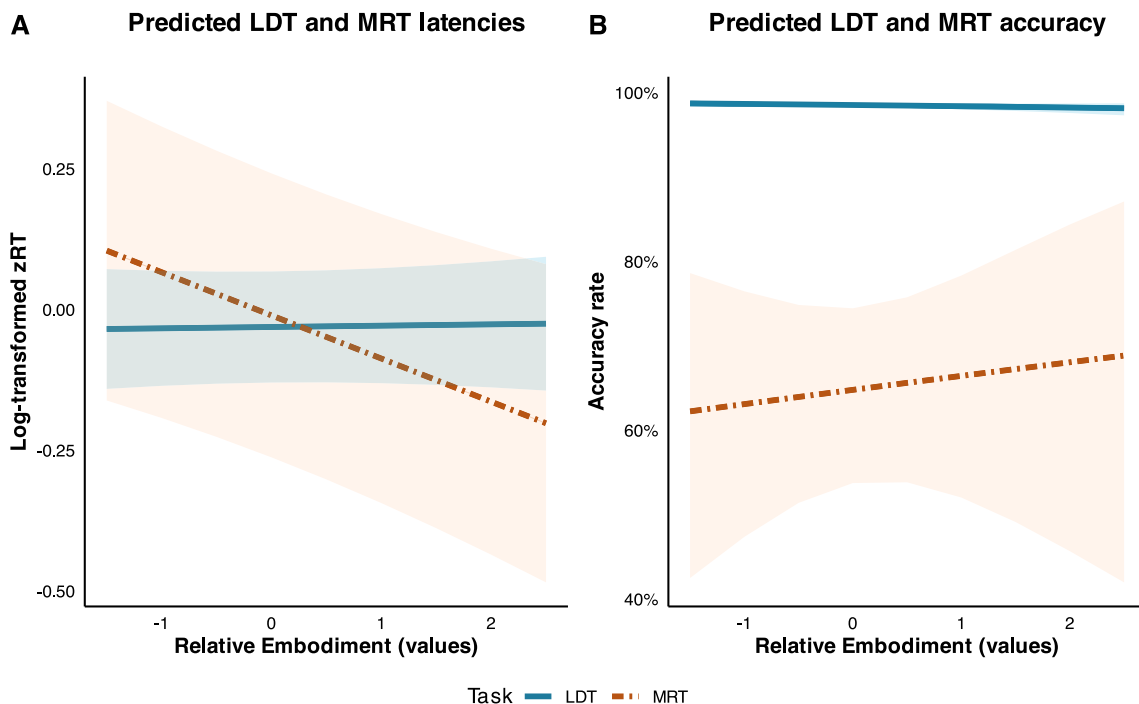


Fig. 5. Results of the cross-task comparison between LDT and MRT. (A) Predicted values of log-transformed z-scored latencies obtained in the Lexical Decision Task (LDT) and the Memory Recognition Task (MRT). (B) Predicted values of accuracy scores obtained in the LDT and the MRT.

The comparison between LDT and MRT log-transformed z -scored latencies (Table 13, Fig. 5A) showed a main effect of the Verbal Meaning Task total score and RE values as well as a significant interaction between RE and Task, indicating that latencies were faster for verbs with higher RE and for individuals with higher vocabulary skills, but the effect of RE was significantly higher in the MRT.

The comparison between LDT and MRT accuracy scores showed no main effect of RE values, yet a significant main negative effect of lexical frequency and word length, indicating that accuracy decreased as word length and lexical frequency increased, as well as a main effect of Task, showing that accuracy levels were significantly lower in MRT (Table 14, Fig. 5B).

Discussion

The results of the MRT showed that, for Italian verbs, RE effects generalize from word processing to memory processes and facilitate word memorization and recognition, thus confirming the prediction based on English data (Sidhu & Pexman, 2016). The across-task comparison is also in line with prediction and confirms the idea, put forward by multiple representation models, that the activation of sensory-motor properties of words varies as a function of task demands. In particular, our results support the greater role of experience-based semantic information in tasks with multiple memory processes involved (e.g., both lexical-semantic and episodic processes).

The results of the MRT also showed a main effect of vocabulary skills (furtherly confirmed by the cross-task comparison) without any interaction with RE levels. This finding suggests that individuals with deeper word knowledge can more efficiently capitalize on their vocabulary experience, but not on sensory-motor properties of words.

General discussion

In this study, we tested the effect of a semantic dimension known as *relative embodiment* (RE) on verb processing in Italian. In particular, we collected RE ratings for a heterogeneous set of Italian verbs – which are offered as a novel, openly accessible resource to be reused for further studies – and to test the modulation of RE effects during verb processing across tasks and at the interface of language and memory. To achieve this latter aim, we considered specifically a task with moderate activation of semantic properties (i.e., lexical decision), a task that in English has shown strong involvement of semantic processing (i.e., grammatical decision), and a task with high semantic demands and additional involvement of other memory processes, such as episodic processes (i.e., memory recognition task). Elaborating on multiple representation theories of word meaning (see Muraki et al., 2023), we made predictions based on data collected from English (Sidhu et al., 2014; Sidhu & Pexman, 2016): in particular, we expected semantic effects to increase across tasks in parallel with semantic demands and episodic memory involvement. Additionally, we also acknowledged potential modulations of RE effects due to Italian-specific characteristics, which might manifest especially in tasks where semantic processing is coupled with grammatical processes (i.e., in grammatical decision).

Overall, our results showed that RE is a relevant dimension of Italian verbs, with RE rating values being strongly associated with ratings acquired for English verbs. Beyond that, the results of the three experiments showed a clear variation of RE effects on verb processing and confirmed that the extent to which RE became available and relevant during verb processing varied across contexts. On the one hand, we observed a significant effect of RE in lexical decision and memory recognition tasks (stronger in the latter compared to the former), indicating that verbs with higher RE values are more easily retrieved from the mental lexicon and more accurately remembered. On the other hand, we did not find similar effects in the grammatical decision task, where verbs with higher RE values were not processed faster than verbs with lower RE values. The pattern that we observed across tasks is in line with

multiple representation theories, which predict that both experience-based and amodal information contribute to shaping the representation of word meaning, in a trade-off that varies as a function of task demands to capitalize on the most salient source of information needed to provide optimal responses (see Connell, 2019; Connell & Lynott, 2013). Notably, the pattern that we observed aligns with predictions from English for two out of three tasks, differing only in the grammatical task. This seems to indicate that task demands interact with language-specific characteristics – such as inflectional richness of grammatical categories – and this interaction drives the activation of either linguistic or experience-based representations.

Focusing on task modulations, lexical decision and memory recognition emerged as the two contexts where RE played a role. These two tasks equally require the activation of experience-based properties of the stimuli yet at a different level of processing. In lexical decisions, we observed that lexical and orthographic properties of the stimuli were more relevant to solving the task, while experience-based properties of verbs were ancillary, thus only indirectly recruited (Yap et al., 2015). This pattern can be interpreted along the lines of models assuming that during shallow lexical operations, semantic representations are not fully activated, but only partially involved via indirect feedback from lexical-orthographic levels (Balota, 1990). Conversely, in tasks requiring full semantic decoding and capitalizing on additional mnemonic processing (as in memory recognition, Ballot et al., 2021; Cortese et al., 2010), we observed that experience-based properties of verbs were highly salient and relevant, even when the effect of lexical variables and imageability controlled. Contrary to previous data on noun processing, where imageability was a better predictor of memory recognition performance than motor-related dimensions (Khanna & Cortese, 2021), our results suggest that the activation of sensory-motor and bodily-related experience is especially relevant for encoding verbs in memory, even beyond amodal properties of words and perceptual-based semantic dimensions.

Results from the memory recognition task offer relevant theoretical implications for the understanding of the language-memory interface and the organization of declarative memory. Originating from Tulving's influential model of declarative memory (Tulving, 1972), the debate in the memory field revolved around whether semantic and episodic processes can be described as distinct submodules, with little interaction (De Brigard et al., 2022). The strong dissociability between episodic and semantic processes has been challenged by empirical evidence accumulating over the years, showing that these two facets of declarative memory, in fact, partially overlap and are intertwined at both behavioral and neural levels (Greenberg & Verfaellie, 2010; Renoult et al., 2019). Far from being decisive in this long-standing debate, our data provide additional support to the idea that semantic processes influence specific episodic processes involved in word recognition: in particular, the encoding of stimuli associated with richer semantic representations promotes stronger storage of those items in episodic memory and better performance in identifying them as familiar in the retrieval phase (Stern & Hasselmo, 2009). This type of semantic-to-episodic prompting effect has also been observed in non-word memory recognition studies: for instance, Hovhannisyan et al. (2021) tested visual object and word recognition, showing that semantic features associated with the conceptual representation of a given object influenced memory performance and contributed to better performance in both word and object retrieval skills. Overall, these data contribute to elucidating which factors – including semantic experience-based dimensions, such as RE – facilitate the process of memorizing certain words and not others. Further studies should explore whether the prompt of bodily-related semantic information to episodic processes also extends to more explicit operations, such as recollecting processes (Stern & Hasselmo, 2009), namely if high-RE verbs are not only easier to recognize as old or new stimuli but also if RE facilitates the process of explicitly recalling items from memory, as in free recall tasks (see Makri & Jarrold, 2021). This would extend to language and word memory pieces of evidence already observed in autobiographical memory recall tasks, where

specific facets of bodily experience – such as interoception – were shown to influence episodic processes (Messina et al., 2022).

Grammatical decision stands out as the task where we found that bodily information plays no role and is where Italian and English results diverged the most. On the one hand, this finding might cast doubts on the role of bodily information in understanding verbs, on the other hand it is fully compatible with the multiple representation account, as it indicates that the context-modulations of experience-based semantic properties of the stimuli might change from language to language and from task to task, especially at the syntactic-semantic interface. Notably, previous studies on the Italian language reported inconsistent patterns concerning the involvement of deeper conceptual features of items when the grammatical decision task was used, which depends on the specific paradigm employed. For instance, verb-noun grammatical decisions in a two-alternative forced-choice paradigm were shown not to require conceptual processing (Postiglione & Laudanna, 2016), thus aligning with our data acquired using a go/no-go task and contrasting with two-alternative forced-choice grammatical decisions on noun gender, which resulted in limited yet detectable effects of word concreteness (Montefinese, Visalli, Angrilli, & Ambrosini, 2024). While data from English-speaking individuals are more in line with the idea that verb-noun class distinction requires deeper access to the semantic-conceptual level (Yap & Pexman, 2016; for a discussion, see Vigliocco et al., 2011; Vinson & Vigliocco, 2002), data from Italian suggest that this might not be the case in a morphologically rich language. A possible explanation for this finding can be related to the rich morphological inventory of the language and how it can influence word recognition. In particular, word stems in Italian are characterized by the presence of class-specific morphemes, such as prefixes typical of denominal verbs (e. g., *ab-braccia* ‘hugs.3SG’, *ap-prezza* ‘appreciates.3SG’), derivational suffix univocally associated with nouns (e.g., *-zione* as in *infe-zione* ‘infection’, *-ezza* as in *ricchezza* ‘richness’) or with verbs (e.g., *-eggia* as in *dann-eggia* ‘damages.3SG’, *-isce* as in *rap-isce* ‘kidnaps.3SG’), as well as allomorphic differences in the word stem of verbs and nouns due to diachronic reasons. According to some theoretical proposals (Baayen & Schreuder, 2000), word retrieval is influenced by the parsing of its morphological representation into constituents (i.e., roots, prefixes, and suffixes) and might be achieved even when access to the whole-word meaning has not been accomplished yet (see Burani & Thornton, 2003), as supported by evidence from psycholinguistic experiments on Italian complex (Crepaldi et al., 2014; Laudanna et al., 1992) and compound words (Arcara et al., 2014). Hence, in Italian, given the many class-specific morphemes, morphological parsing might be sufficient to accomplish verb-noun class assignments without deep access to the semantic representation, including experience-based dimensions such as Relative Embodiment.

Beyond these considerations, our findings are consistent with some previous – albeit still scarce – evidence of cross-linguistic variation of sensory-motor effects in language processing. For instance, Gianelli et al. (2017) compared the effect of verbs combined with pronouns on the activation of motion-related representations in Italian and German. The presence of the 2SG personal pronoun (i.e., “you”) as the subject of either action or interaction verbs, triggered a motor effect in Italian but not in German. This effect was interpreted in light of structural differences between Italian and German languages: the non-mandatory status of pronouns in Italian might have promoted perspective taking and the re-activation of motor experience during word processing. These data bring additional implications at the methodological level: when it comes to studying the role of the semantic representation of words in morphologically rich languages, such as those within the Romance family, it may be important to consider the interaction between the morphology-semantics interface and structural language-specific differences (Bambini & Canal, 2021), especially when employing a task involving decisions on grammatical properties of the items.

An additional point worth discussing is the role of individual differences in verb processing and their interaction with RE. In this study,

we tested the role of vocabulary and mental imagery skills in an explorative fashion, capitalizing on initial evidence that individual differences interact with semantic processes alongside task and contextual differences (Ibáñez et al., 2023; Pexman, 2020). Our results confirm that individual differences matter in Italian verb processing, highlighting especially the role of between-subject variability at the level of vocabulary skills: participants with deeper vocabulary knowledge and higher lexical quality could capitalize on more extended and robust word knowledge, showing better performance compared to less skilled participants in all three tasks. However, this vocabulary advantage showed remarkable task-related differences, as it directly interacted with bodily properties of the stimuli only in one out of three tasks. In particular, high vocabulary skills resulted in less sensitivity to semantic properties in the lexical decision task but overall faster grammatical decisions and memory recognitions, suggesting a quite broad effect of this competence. Our data confirm patterns already observed in previous studies on the English language, showing that individual differences in vocabulary skills interact with experience-based semantic dimensions in lexical decisions (Pexman & Yap, 2018; Yap, Pexman, et al., 2012) and facilitate grammatical decisions (Muraki & Pexman, 2021), but our findings also provide new evidence on the role of vocabulary in word memory recognition.

Conversely, we did not find an effect of mental imagery in any of the three tasks. Previous studies in English showed that, while mental imagery does not seem to affect lexical decisions, it interacts at least with body-object interaction properties during grammatical decisions (Muraki & Pexman, 2021). Here we did not find such effects, which might be due to the fact that – especially in single-word paradigms – the processes underlying mental imagery might diverge from embodied simulations involved in semantic processing (Meteyard et al., 2012; Willems et al., 2010). It is also important to underline that our analysis only addressed the vividness of first-person, third-person, and kin-aesthetic motor imagery via a self-administered questionnaire: as pointed out by recent viewpoints (see, for instance, Schwarzkopf, 2024), such questionnaires might not capture imagery skills with high levels of precision as a combination of implicit tasks would do. Nevertheless, as observed by Muraki et al. (2023), the investigation of mental imagery is a promising avenue to unravel the embodied simulation processes underlying conceptual knowledge.

Finally, there are certainly some limitations in our study, which might inspire future investigations on the effects of RE in language and memory tasks. The most relevant limitation concerns the fact that we could compare the effects of RE between the lexical decision and the memory recognition tasks only in a subset of items used in both tasks and using responses from two unbalanced samples of participants. This limitation affects the power of the post-hoc analysis, which nevertheless can inform us concerning the different weights of RE in these two tasks. Interestingly, most of the studies comparing semantic effects across tasks limited their observations to qualitative inspections of differences in effect sizes (see, for instance, Khanna & Cortese, 2021): our study, albeit less robust in terms of power, may inspire more direct comparisons of semantic effects across tasks in future studies. Similarly, we approached the role of individual differences in vocabulary and motor mental imagery skills only in an explorative fashion, as sample size across experiments was determined to be powered enough to detect effects involving item-level predictors, but not individual difference variables. This limitation reflects the fact that individual differences are still marginally addressed in the literature investigating semantic richness effects, despite their potential relevance to testing the predictions of multiple representation accounts (Muraki, Dahm, et al., 2023; Muraki & Pexman, 2021): future studies should exploit more robustly their role in modulating semantic effects in word processing to deepen our understanding of the processes underlying their interaction with task properties, contexts, and languages.

Overall, our work documents consistent effects of RE across languages and contributes to enriching the available evidence supporting

the idea that richer semantic representations facilitate word processing and memorizing. Moreover, our data show that these effects are flexible, as they not only seem to vary as a function of semantic demands along the language-memory *continuum* but they become negligible when task demands interact with language-specific grammatical features. Beyond these findings, the semantic richness effects described in our work emphasized also that different memory components (e.g., semantic and episodic processes) likely interact at the language-memory interface: this piece of evidence is expected to significantly contribute to the current theoretical debate on the organization of declarative memory and its relationship with language ability.

CRedit authorship contribution statement

Federico Frau: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Luca Bischetti:** Writing – original draft, Methodology, Formal analysis. **Lorenzo Campidelli:** Investigation. **Elisabetta Tonini:** Software, Methodology. **Emiko J. Muraki:** Writing – review & editing, Methodology. **Penny M. Pexman:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Valentina Bambini:** Writing – original draft, Supervision, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data file and analysis codes for the present study are available through the Open Science Framework (OSF) at the following link: <https://osf.io/axhn3/>.

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